MILITARY SPECIFICATION

VALVE, FLUSH (FLUSHOMETER), WATER CLOSET, URINAL, AND SERVICE SINK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. Scope. This specification covers requirements for flush valves (flushometers), fittings, and trim for land use.

1.2 Classification.

1.2.1 Definitive part number (DPN). Items furnished in accordance with this specification shall be identified by a DPN. This number is intended for cataloging and ordering purposes; it is not intended for surface printing on items in accordance with this specification. The DPN shall consist of the Military specification symbol plus the specification part number (SPN) (see 1.2.2). The DPN shall be written as indicated below (see 6.2 and 6.3):

DPN M29193 XXXX

Military specification symbol

SPN (see 1.2.2)

1.2.2 Specification part number. The SPN shall be a three position numeric and a one position alpha code which identifies the flushometer in accordance with this specification (see Table I).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4510

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TABLE I. DPN identification.

SPN	Characteristic
1	Exposed installation.
2	Concealed installation.
a	Handle operated.
b	Pedal operated.
c	Push button operated.
d	Push disk operated.
e	Solenoid operated.
1.5	1-1/2 gallons per flush (for urinals).
3.5	3-1/2 gallons per flush (for water closets and service sinks).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-P-12808 - Plumbing Fixtures and Accessories, Packaging of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

No. 73 - Motor-Operated Appliances.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

3. REQUIREMENTS

3.1 Standard commercial product. The flushometers shall, as a minimum, be in accordance with the requirements of this specification and shall be the

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manufacturer's standard commercial product with any added features needed to comply with the requirements. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product shall be included in the flushometers being furnished. Standard commercial product is a product which has been sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

3.2 Identical items. Flushometers of the same classification furnished under any specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable without prior written approval of the contracting officer.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specifically specified.

3.3.1 Flushometers. The flushometers, fittings, and trim shall be copper alloy or stainless steel, at the manufacturers option. Internal parts such as, diaphragm, gaskets, and O-rings, shall be leather, rubber, or any suitable elastomeric material for the intended purpose.

3.3.1.2 Dissimilar metals. Immediate contact which can be expected to cause galvanic corrosion shall be avoided. When such contact cannot be avoided interposing insulating material shall be provided to minimize the corrosive effect.

3.4 Design and construction.

3.4.1 Flushometers. Each flushometer shall include the necessary fittings and trim for complete installation from the fixture to the water supply line, and as a minimum shall be the manufacturer's standard commercial product.

3.4.1.2 Solenoid flushing device. Flushing device shall be solenoid hydraulic operated. Each contractor shall submit to the contracting agency, proof that the solenoid he proposes to supply in connection with the flushometer conforms to the requirements of UL No. 73. The label, or listing with re-examination of the UL, may be accepted as evidence that the solenoid conforms to this requirement. In lieu of the label, or listing with re-examination, the contractor may submit independent proof, satisfactory to the contracting agency, that the solenoid conforms to the applicable requirements of UL No. 73.

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3.4.2 Operation. The flushometers shall be capable of continuous reliable operation without sticking, chattering, or causing water hammer (see 4.6.2).

3.4.3 Ease of maintenance. The flushometer shall be so designed and constructed that repairs and overhaul can be readily accomplished by means of general purpose tools with a minimum removal of disturbance of other elements of the unit.

3.4.3.1 Volume. Unless otherwise specified (see 6.2 and 6.4), the flushometer shall deliver 1.5 ±.025 gallons of water for urinals and 3.5 ±.500 gallons of water for water closets and service sinks (see 4.6.2 and 6.2).

3.4.4 Controls stop valve. When specified (see 6.2), control stop valve shall include a check valve.

3.5 Finish. Exposed type valves, fittings, and trim shall be bright chromium plated in accordance with the manufacturer's current standard practice. Concealed valves and fittings may be plated or unplated at the manufacturers option.

3.6 Marking. Each valve shall be permanently marked with the trade mark or name of the manufacturer.

3.7 Workmanship. Flushometers shall be free of defects which affect serviceability or operation. Metals and plastics in flushometers shall be free from cracks and defects which impair serviceability or life.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.2.1).
(b) Packaging inspection (see 4.7).

4.2.1 Quality conformance inspection. Quality conformance inspection shall be performed on the sample valves selected in accordance with Section 4. This shall include the examination of 4.5 and the tests of 4.6.

4.3 Inspection lot. All flushometers of the same parts, types, styles, and classes offered to the Government at one time, shall be considered a lot for inspection. The sample unit shall be one complete flushometer.

4.4 Sampling. A random sample of valves shall be selected from each lot in accordance with MIL-STD-105.

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4.4.1 Sampling for examination. Examination of the valves shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.4.2 Sampling for test. Tests of the valves shall be based on an inspection level S-2 and an AQL of 4.0 percent defective.

4.5 Examination. Each sample selected in accordance with 4.4.1 shall be examined for compliance with the requirements of Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This inspection shall encompass all visual examinations and dimensional measurements.

4.6 Tests. Each sample selected in accordance with 4.4.2 shall be tested as specified herein.

4.6.1 Solenoid flushing device. Test solenoid flushing device in accordance with UL No. 73 (see 3.4.1.2).

4.6.2 Operational. Adjust the flushometer control stop valve for maximum flow. At an inlet pressure of 50 ±3 pounds per square inch (psi) operate the valve five times and measure the volume of water discharged at each operation. Perform the same test at an inlet pressure of 20 psi (see 3.4.2 and 3.4.3.1).

4.6.2.1 Hydrostatic. Test the flushometer at an inlet pressure of 100 ±5 psi with the valve body plugged at openings for strength, porisity, and leakage.

4.6.2.2 Leakage. Adjust the stop for maximum flow, then test the flushometer at 50 psi minimum inlet pressure. There shall show no leakage.

4.7 Packaging inspection. The preservation-packaging, packing, and marking of the flushometers shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation-packaging, packing, and marking. The preservation-packaging, packing, and marking shall be in accordance with MIL-P-12808 except, unless otherwise specified (see 6.2), a preproduction pack shall not be required. The level of preservation-packaging and the level of packing shall be as specified (see 6.2).

6. NOTES

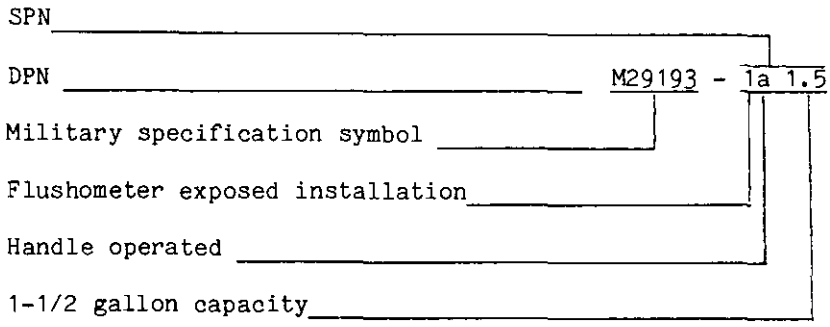
6.1 Intended use. Flushometers are intended for flushing wastes from plumbing fixtures into sewer systems.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

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(a) Title, number, and date of this specification.
(b) Definitive part number (see 1.2.1 and 6.3).
(c) Volume requirements (see 3.4.3.1 and 6.4).
(d) When check valve is required (see 3.4.4).
(e) Level of preservation-packaging, and level of packing required (see 5.1).
(f) If a preproduction pack is required (see 5.1).

6.3 DPN example. The letter "M" is used for Military specification and the Military specification No., without the letter "V" (valve). Combined they form the definitive part No.



6.4 Water pressure and volume requirements. Purchasers should note that these 1.5 and 3.5 gallon flushometers are the most common types. They will operate on a wide pressure range (15 to 50 psi while flushing) without significantly effecting the delivered volume. Volume requirements are determined by the fixture being flushed, not the flushometer. Most new fixture installations require a volume of 3.5 ±.5 gallons for water closets and 1.5 ±.025 gallons for urinals. These limits may vary from one area to another. Some older fixtures may require even higher volumes of water (6 to 7 gallons). Purchasers should determine volume requirements and consult manufacturer's literature prior to ordering flushometers.

6.5 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Aquisition Regulation (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing activity:

Navy - YD

Project No. 4510-0172

Review activity:

DLA - CS

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-V-29193 - Valve, Flush (Flushometer), Water Closet, Urinal and Service Sink

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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Replaces edition of 1 Jan 72 which may be used.

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